

SupportDesk

Business Problem & Solution Definition

BUSINESS PROBLEM Fragmented internal support requests make it difficult to track issues, conversations, ownership and resolution status.	PROPOSED SOLUTION A centralized operational support application using Databricks Apps and Lakebase.	BUSINESS OUTCOME A reliable source of truth for support operations and a foundation for future AI capabilities.
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1. Business Context

Organizations regularly receive internal support requests from employees. Examples include application login failures, dashboard issues, data refresh failures, access requests, and incorrect reports. These requests are often communicated through email, chat applications, spreadsheets, or informal conversations, making it difficult for support teams to maintain a single, reliable view of each issue.

2. Business Problem

The company needs a centralized operational support system that allows employees and support teams to create support tickets, maintain the conversation history for each ticket, track the current status of an issue, and update tickets throughout their resolution lifecycle.

Without a dedicated operational system, support information can become fragmented. Support teams may struggle to identify which issues are still open, what actions have already been taken, who is responsible for an issue, and whether a ticket has been resolved. This can lead to slower resolution times and a poorer employee support experience.

Problem statement: The company needs a centralized operational support system to track employee issues, preserve their conversation history, and manage each issue through its resolution lifecycle.

3. Example Business Scenario

An employee is unable to access the company's sales dashboard. The employee creates a support ticket describing the issue. A support engineer investigates the problem and adds messages to the ticket. The ticket moves through a simple lifecycle: **Open** → **In Progress** → **Resolved**.

Business Event	Information Captured
Employee reports an issue	Ticket title, creator, creation time and status

Business Event	Information Captured
Support team responds	Message, author, timestamp and related ticket
Investigation begins	Ticket status changes to In Progress
Issue is fixed	Ticket status changes to Resolved

4. Business Objectives

- Centralize internal support requests in one operational system.
- Maintain a complete conversation history for each support ticket.
- Track the lifecycle of every issue from creation to resolution.
- Allow support teams to update tickets and respond quickly.
- Provide reliable operational data that can later support analytics and AI-powered capabilities.

5. Proposed Solution

Build a small internal support application using **Databricks Apps** as the application layer and **Lakebase** as the operational database. Users can view existing tickets, select a ticket to view its messages, create new tickets, add messages, and update ticket status.



6. Why Lakebase?

The application performs transactional operations such as creating individual tickets, adding messages, retrieving messages for a selected ticket, and updating ticket status. These are operational database workloads rather than purely analytical workloads.

Lakebase provides a PostgreSQL-based operational data layer that can support these application transactions while remaining within the Databricks ecosystem. This project therefore demonstrates the distinction between operational application data and analytical data.

7. Core Data Model

Table	Purpose	Key Relationship
tickets	Stores the main support request and its current status.	ticket_id is the primary key.
ticket_messages	Stores the conversation history for each ticket.	ticket_id references tickets.ticket_id.

8. Business Process

1. An employee or support user creates a ticket.
2. The ticket is stored in Lakebase with an initial status such as Open.
3. Users can add messages to the ticket as the issue is investigated.
4. The support team updates the ticket status as work progresses.
5. The issue is eventually marked Resolved.
6. The complete ticket and message history remains available for future reference.

9. Expected Business Value

- Better visibility into current support issues.
- Improved traceability of support conversations.
- A consistent process for tracking issue resolution.
- A reliable operational source of truth for the support application.
- A foundation for future reporting, automation, context engineering, and AI support agents.

10. Future AI Opportunity

The ticket and message history created by this application can become a foundation for future AI capabilities. An AI support assistant could summarize conversations, identify the current issue, retrieve relevant historical context, recommend troubleshooting steps, and help support engineers decide what action to take next.

11. Project Success Criteria

- Users can view support tickets stored in Lakebase.
- Users can view messages belonging to a selected ticket.
- Users can create a new ticket.
- Users can add a message to an existing ticket.
- Users can update ticket status.
- Changes persist in Lakebase after the application is refreshed.

12. Project Scope

This Day 1 project intentionally focuses on the core operational workflow. Advanced capabilities such as authentication, AI agents, RAG, notifications, analytics pipelines, and production-grade observability can be added in later iterations.